



Training Program 1: Advanced NLP & Large Language Models (LLMs)

- M01: Text Preprocessing
- M02: Word and Sentence Embeddings
- M03: Transformer Architectures
- M04: Entity-Centric NLP
- M05: Applied NLP Tasks
- M06: Multimodal NLP
- M07: LLM Fundamentals
- M08: Prompt Engineering
- M09: Fine-Tuning & PEFT
- M10: RAG & Real-World Projects



Advanced NLP & Large Language Models (LLMs)

10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

10.2 RAG Architectures: Retrieval-Augmented Generation Systems

10.3 Real-World Applications of LLMs and RAG

10.4 Presentation Guidelines and Final Review

10.5 Student Capstone Project Presentations

10.6 Course Wrap-up and Next Steps

M10 RAG & Real-World Projects

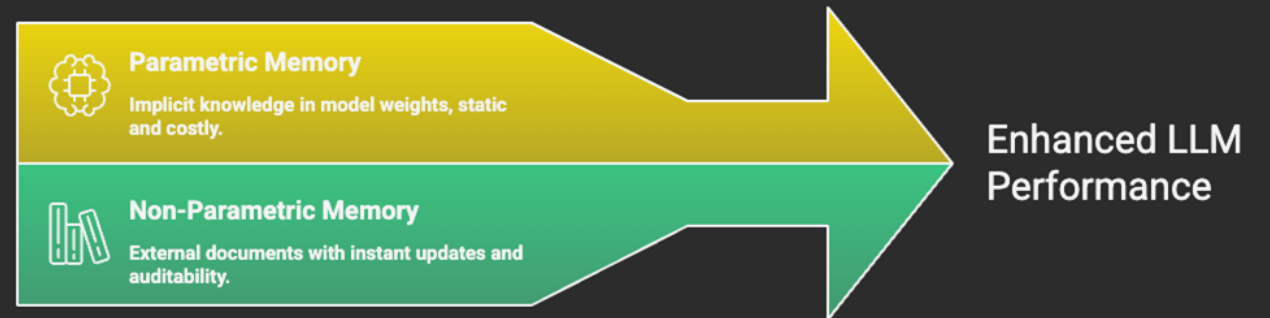
Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

Why RAG Exists

Large Language Models are powerful reasoning engines, but their knowledge is frozen in time. Retrieval-Augmented Generation (RAG) solves this limitation by connecting models to real, up-to-date, and private information sources.

- LLMs suffer from a knowledge cut-off
- They cannot access private or proprietary data
- Fine-tuning is not designed to inject new facts
- RAG enables real-time knowledge access without retraining

Memory Strategies for LLMs



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Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

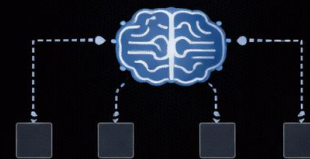
Parametric vs Non-Parametric Memory

LLMs store knowledge implicitly in weights, while RAG introduces explicit, searchable memory. This distinction is critical for reliability, updates, and compliance.

- Parametric memory: knowledge encoded in model weights
- Static and expensive to update
- Non-parametric memory: external documents + retrieval
- Instantly updateable and auditable

Parametric vs Non-Parametric Memory

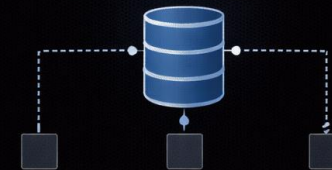
Parametric Memory



Fixed weights

In-memory knowledge
from fixed model weights

Non-Parametric Memory



External data

On-demand knowledge
from external sources

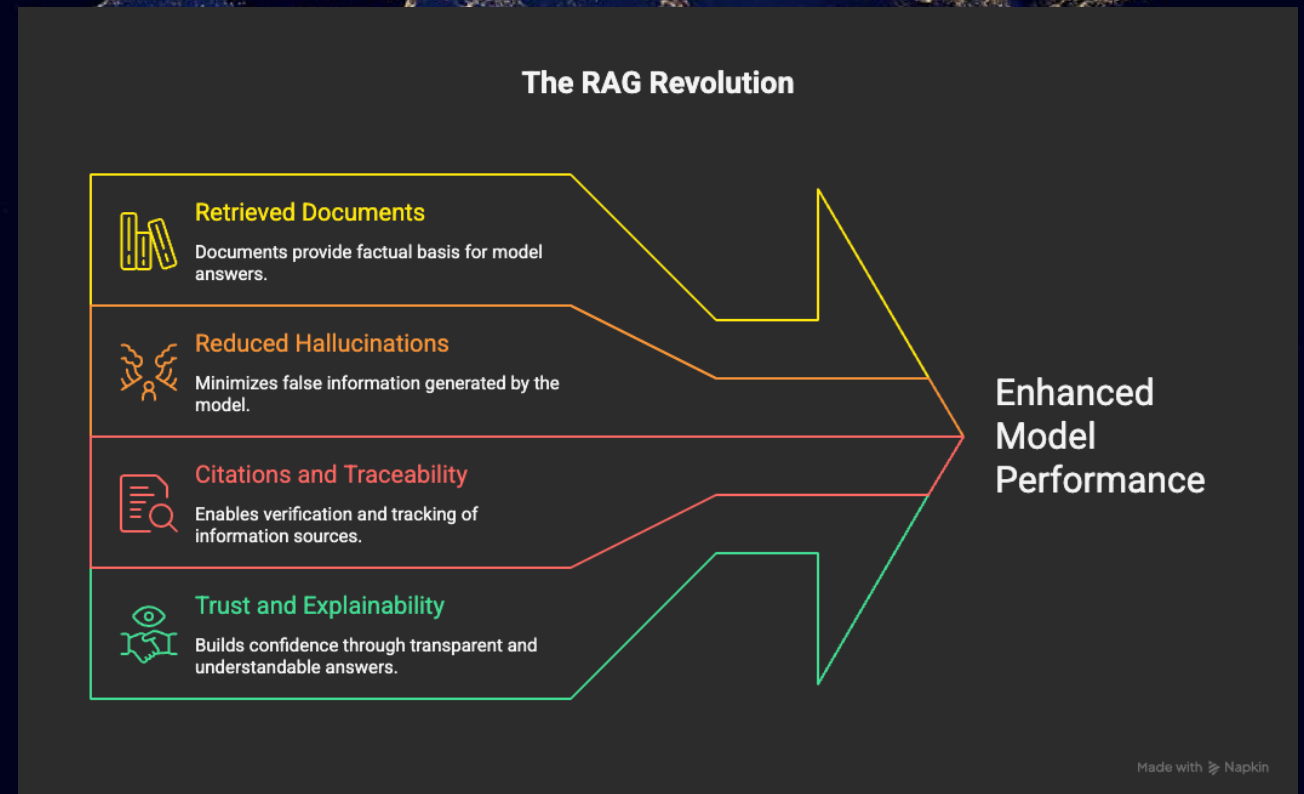
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Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

What Retrieval-Augmented Generation Is

RAG transforms the model from a “knower” into a “reader”, forcing answers to be grounded in retrieved documents visible in the context window.

- Model answers using retrieved documents
- Reduces hallucinations
- Enables citations and traceability
- Improves trust and explainability



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Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

RAG vs Fine-Tuning

RAG and fine-tuning solve different problems. Confusing them leads to brittle systems. The strongest architectures use both strategically.

- Fine-tuning: style, tone, domain behavior
- RAG: factual knowledge and updates
- RAG avoids catastrophic forgetting
- Enterprise systems often combine PEFT + RAG

Fine-Tuning vs RAG

	Fine-Tuning	RAG
Purpose	Style, tone, domain behavior	Factual knowledge, updates
Cost	Higher: requires retraining	Lower: uses existing data
Update Speed	Slower	Faster
Risk	Catastrophic forgetting	Brittle retrieval failures

Enterprise AI Decision





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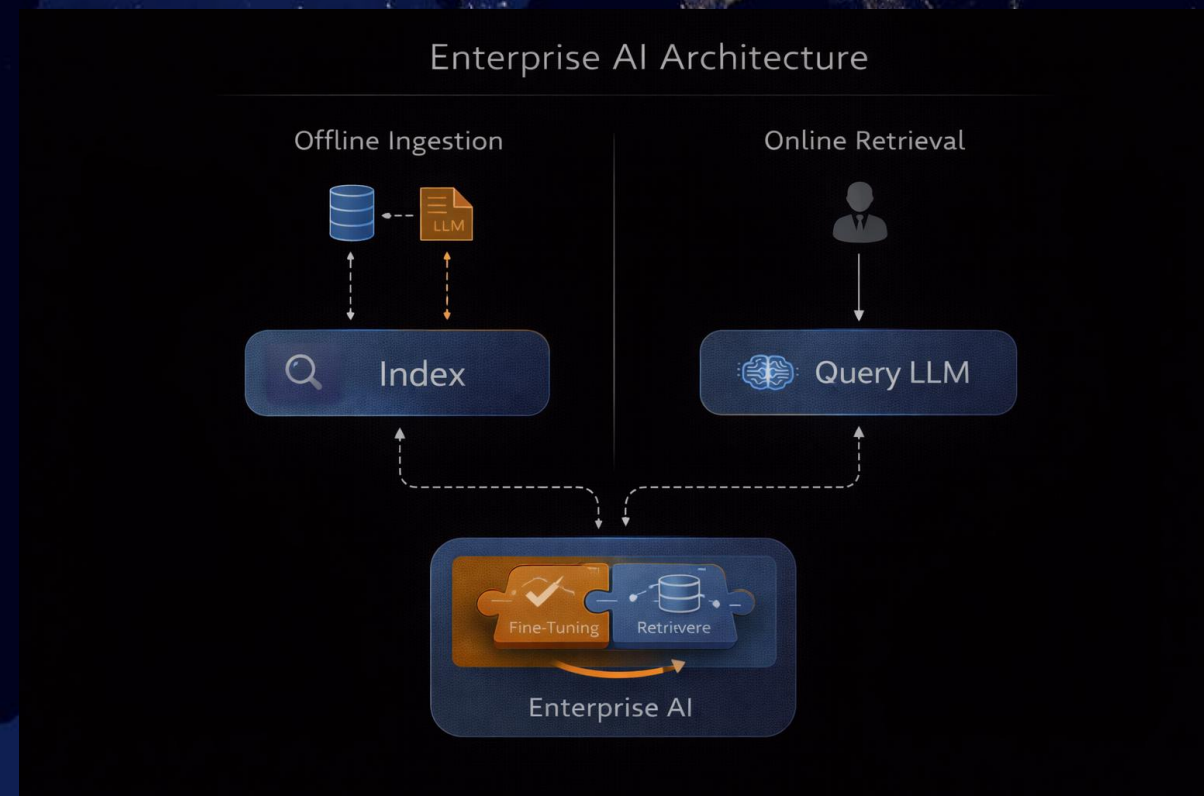
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Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

The Hallucination Problem

LLMs are optimized for fluency, not truth. Hallucinations occur when the model fills gaps probabilistically instead of relying on verified information.

- Fluent $\neq$ factual
- Hallucinations scale with confidence
- Dangerous in legal, medical, and financial domains
- Retrieval acts as a grounding constraint



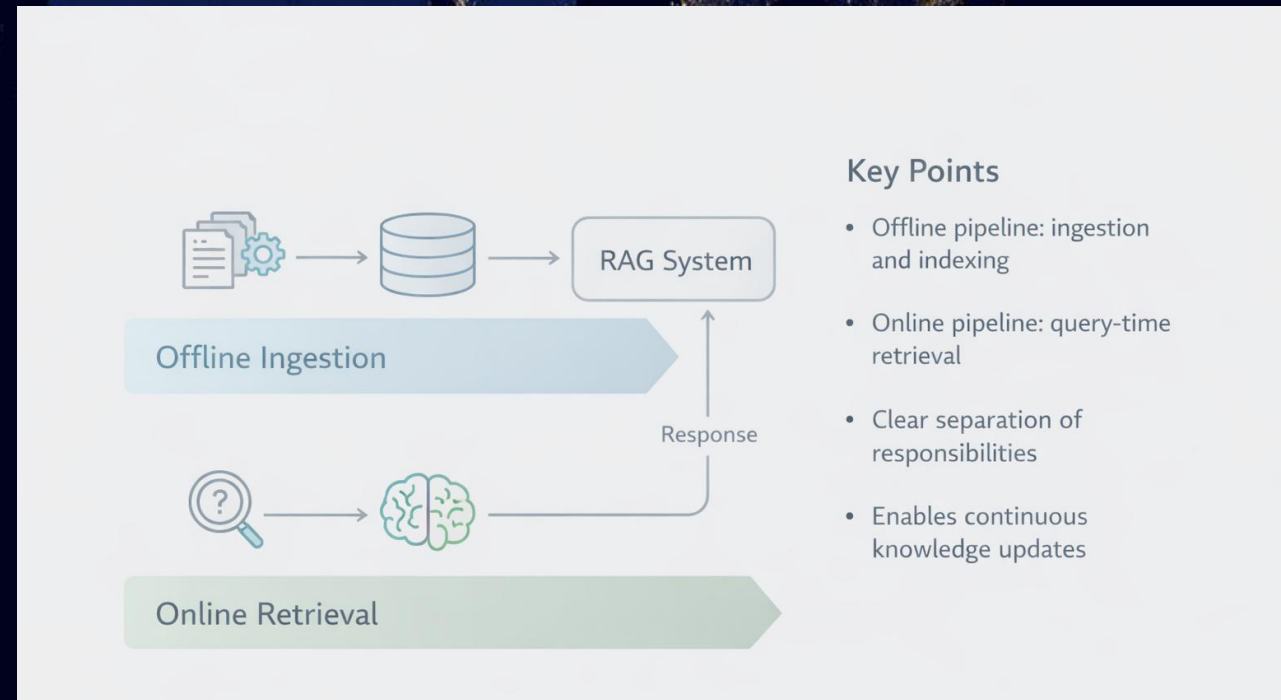
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Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

The Dual-Pipeline RAG Architecture

A production RAG system separates knowledge preparation from real-time reasoning to ensure scalability and performance.

- Offline pipeline: ingestion and indexing
- Online pipeline: query-time retrieval
- Clear separation of responsibilities
- Enables continuous knowledge updates



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Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

Indexing Pipeline: Chunking & Embeddings



Playing

Chunking Chunks

Vector
Transformation

Key Points

- Chunking documents into semantic units
- Overlapping chunks improve recall
- Embedding models convert text into vectors
- Stored in vector databases (FAISS, Pinecone, Milvus)

Before retrieval is possible, documents must be transformed into searchable vector representations.

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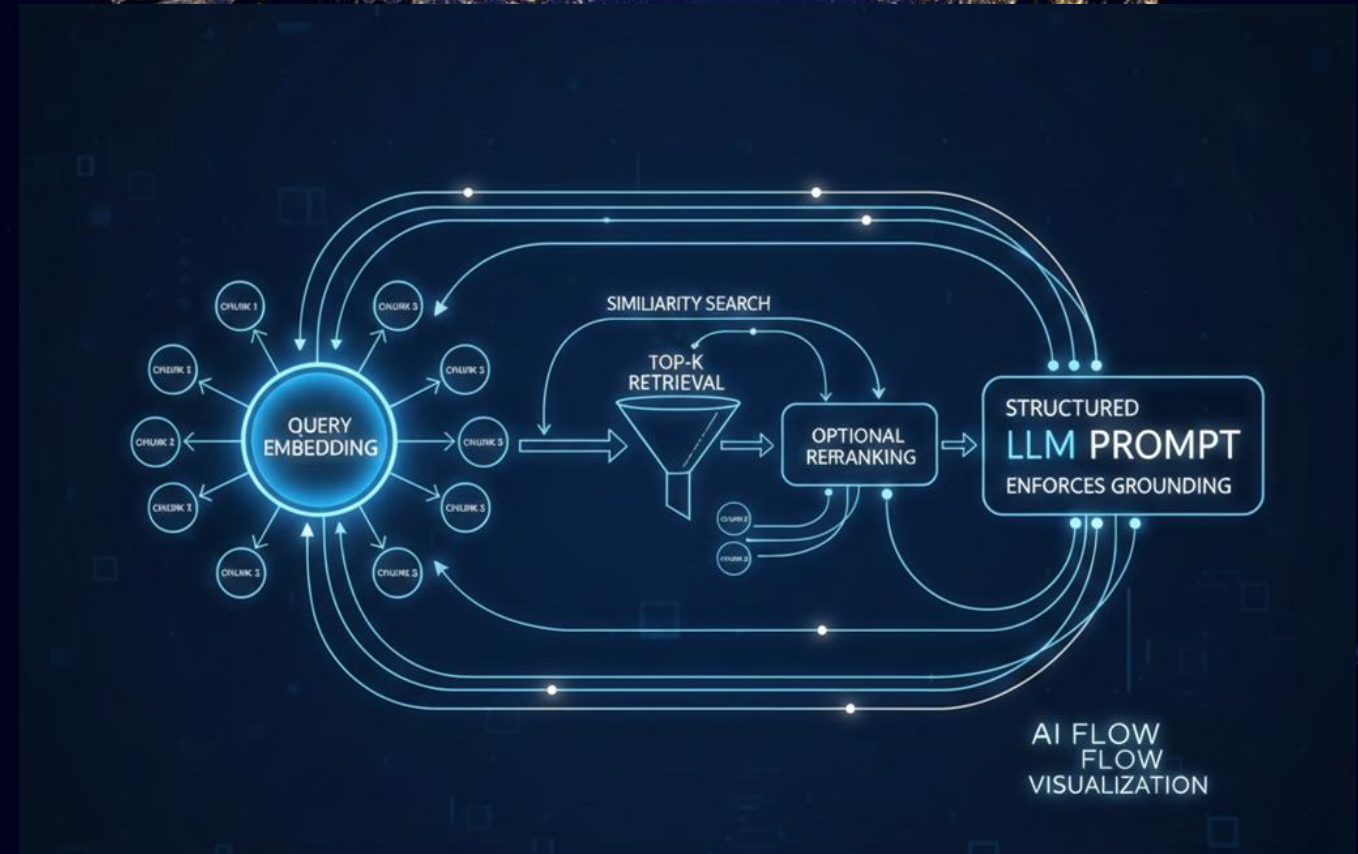
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Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

Retrieval and Prompt Augmentation

At query time, the system retrieves the most relevant chunks and injects them into the prompt to guide generation.

- Query embedding + similarity search
- Top-K retrieval
- Optional reranking for precision
- Structured prompt enforces grounding





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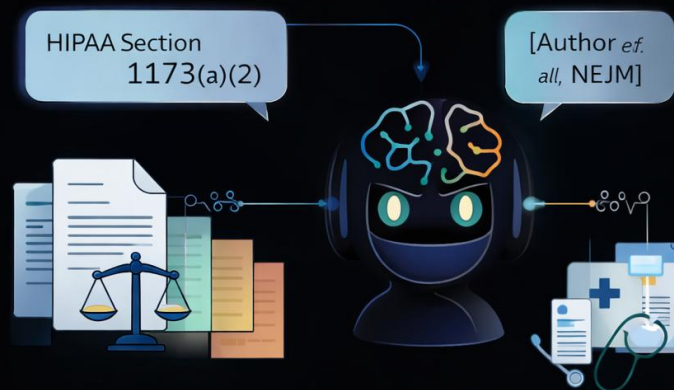
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Chapter 10.3 Real-World Applications of LLMs and RAG

Legal & Healthcare Use Cases

Legal & Healthcare Use Cases

Summary: RAG enables high-stakes decision support where accuracy, traceability, and compliance are mandatory.



Key Points

- Legal: contract review, regulation audits
- Healthcare: clinical decision support
- Answers backed by source documents
- Reduced manual review time

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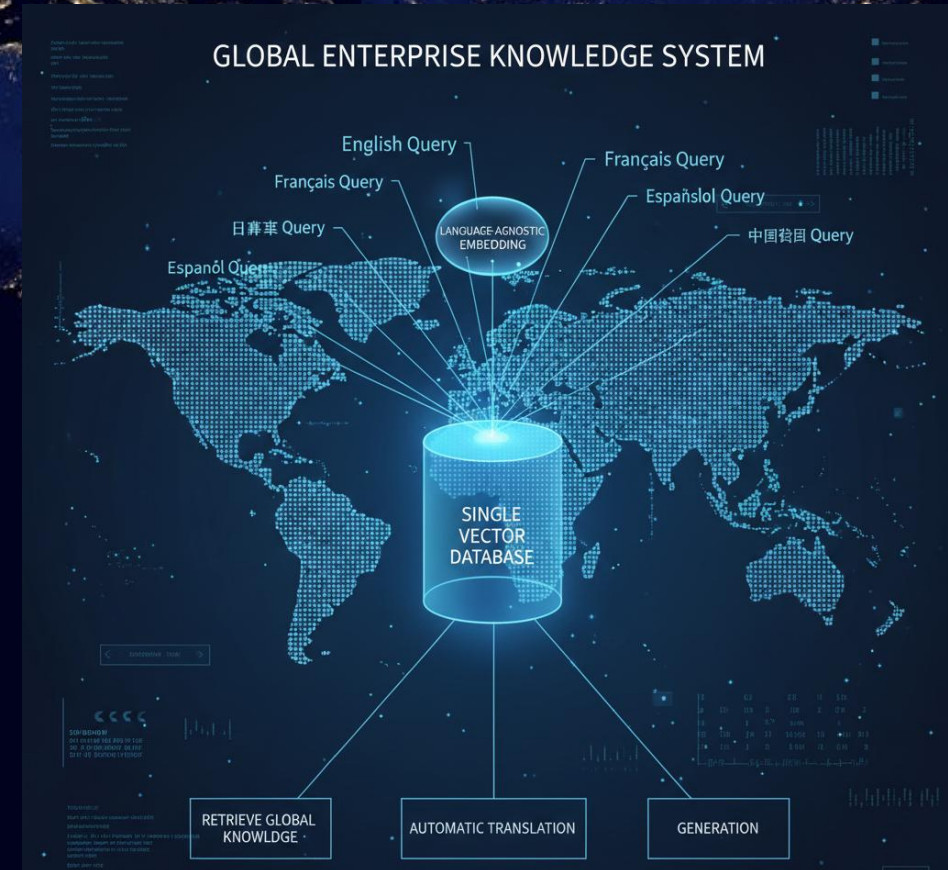
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Chapter 10.3 Real-World Applications of LLMs and RAG

Multilingual Enterprise Knowledge

Modern embeddings allow organizations to access global knowledge regardless of language barriers.

- Language-agnostic embeddings
- Query in one language, retrieve in another
- Centralized global knowledge
- Automatic translation at generation time





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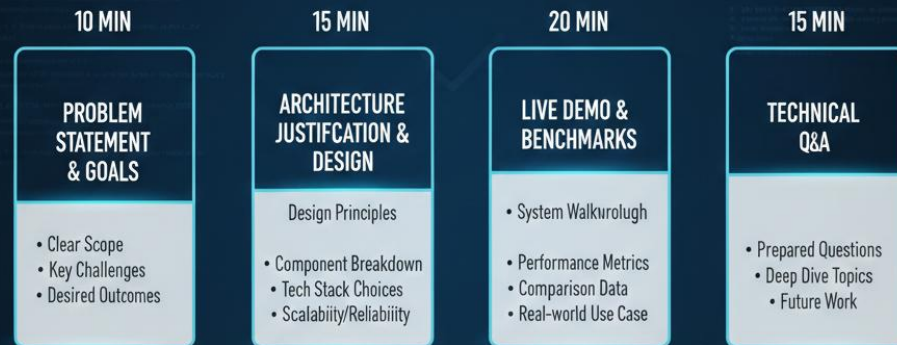
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Chapter 10.4 Presentation Guidelines and Final Review

How to Present a RAG System

AI ARCHITECTURE DEFENSE: EVIDENCE-BASED DECISIONS



TOTAL: 60 MINUTES — DATA-DRIVEN VALIDATION

A professional AI presentation defends architectural decisions with evidence, not narratives.

- Clear problem statement
- Architecture justification
- Live demo or benchmarks
- Prepared technical Q&A

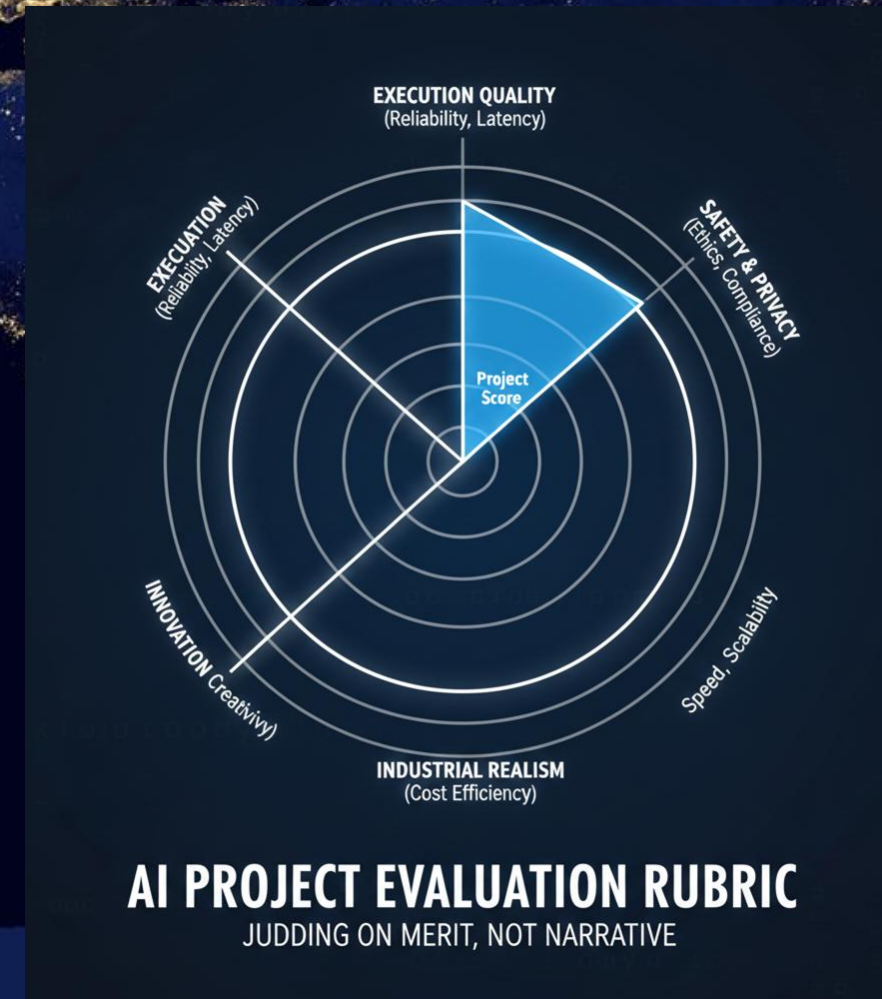
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Evaluation Criteria

Projects are judged on execution quality, innovation, and industrial realism.

- Retrieval quality
- Latency and scalability
- Safety and privacy
- Cost efficiency



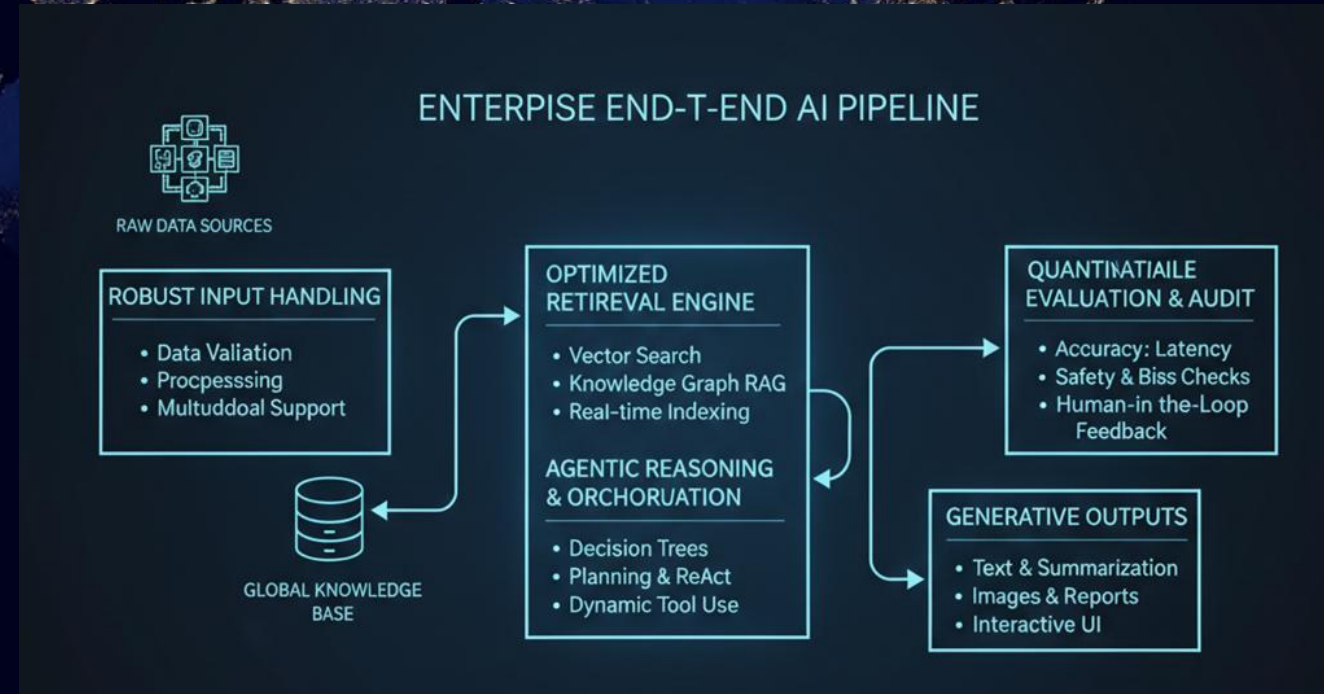
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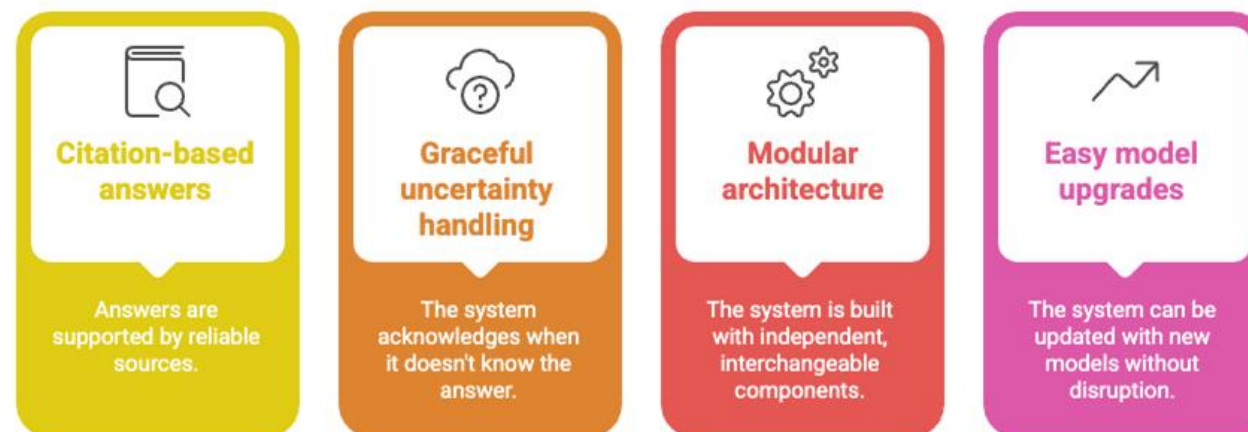
Chapter 10.5 Student Capstone Project Presentations

What Excellence Looks Like

Capstone projects demonstrate full-stack AI system thinking, not isolated techniques.

- Robust input handling
- Optimized retrieval
- Agentic reasoning
- Quantitative evaluation

Project Characteristics

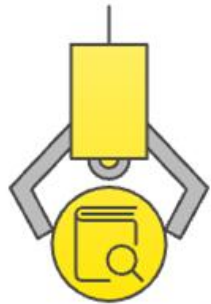


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Chapter 10.5 Student Capstone Project Presentations

Production-Ready Criteria

AI project features



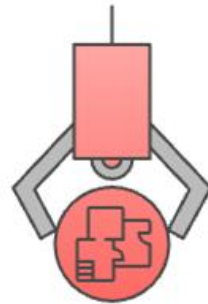
Citation-based answers

Answers are supported by citations for verification.



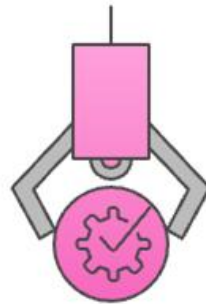
Graceful uncertainty handling

The system handles uncertainty in a user-friendly manner.



Modular architecture

The system is built with modular components for flexibility.



Easy model upgrades

The system allows for simple and efficient model upgrades.

Top projects balance factuality, usability, and maintainability.

- Citation-based answers
- Graceful uncertainty handling
- Modular architecture
- Easy model upgrades



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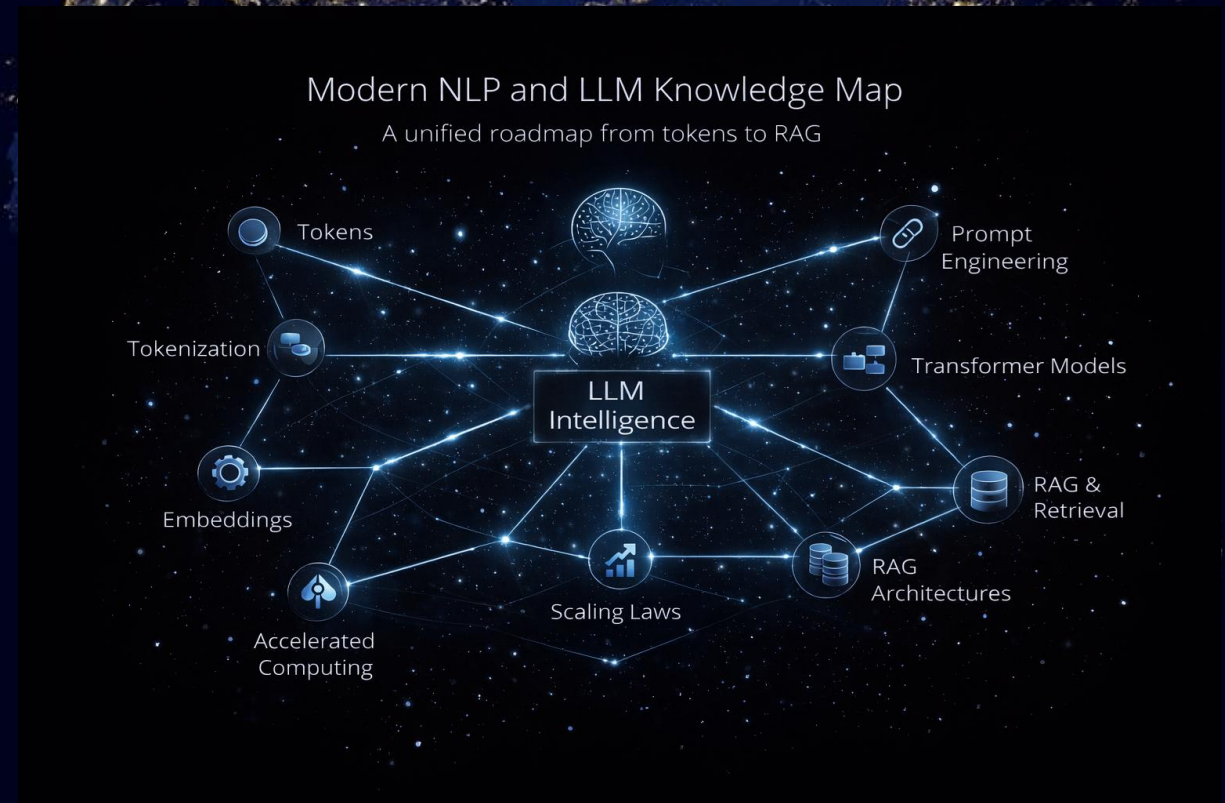
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Chapter 10.6 Course Wrap-up and Next Steps

What You Have Mastered

This course covered the complete modern NLP and LLM stack, from tokens to enterprise-grade systems.

- Tokenization & embeddings
- Transformers & scaling laws
- Prompt engineering
- PEFT & LoRA
- RAG architectures



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Chapter 10.6 Course Wrap-up and Next Steps

The Road Ahead

The future of AI lies in grounded, autonomous, and agentic systems that act, not just respond.

Key Points

- Agentic workflows
- Multimodal reasoning
- Autonomous task execution
- Continuous learning mindset



Tools



Web



Databases



LLM
Intelligence

The future of AI lies in grounded, autonomous, and agentic systems that act, not just respond.

- Agentic workflows
- Multimodal reasoning
- Autonomous task execution
- Continuous learning mindset

M10 RAG & Real-World Projects

References & Further Reading

- Lewis et al. (2020) — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
- Guu et al. (2020) — REALM: Retrieval-Augmented Language Model Pre-training
- Borgeaud et al. (2022) — Improving Language Models by Retrieving from Trillions of Tokens (RETRO)
- Mialon et al. (2023) — Augmented Language Models: A Survey
- Es et al. (2023) — RAGAS: Automated Evaluation of Retrieval-Augmented Generation
- Barnett et al. (2024) — Seven Failure Points When Engineering a RAG System
- Zobel & Moffat (2006) — Exploring the Boundaries of Low-Cost Retrieval Evaluation
- Bender et al. (2021) — On the Dangers of Stochastic Parrots